

The (active) MATRIX

LCD monitors: a look behind the screens

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Flat panel technology has slowly evolved from just a niche cutting edge display solution to become a serious contender for the mainstream display market. It has progressed significantly from dull passive matrix screens to vibrant and fast-responding active matrix screens. The simplicity of the passive matrix system was beautiful, but it had significant drawbacks, notably slow response time and imprecise voltage control. Active matrix LCDs, which have faster pixel response time, depend on thin film transistors (TFT).

It is difficult to manufacture a TFT as compared to the ordinary CRT, primarily because the display size is limited by the number of transistors. As the number of transistors increase, so does the possibility of incorporating a bad or dead transistor. These show up as black dots on the screen, more popularly known as a 'bad pixels'. Most active matrix displays have a few bad pixels scattered across the screen, which is extremely irritating and is the primary reason why manufacturers tend to concentrate on smaller sizes. Manufacturers of large TFTs can go as far as to reject up to 50 per cent of the panels that are manufactured. Given such a low rate of yield, the prices of flat panels are quite high. Despite the lower yields on larger screen sizes, the best advantage is that

What Lights Beneath?

TFTs (Thin Film Transistors) are tiny switching transistors arranged like a matrix on a glass substrate. To light up a particular pixel, the proper row and column are switched on, and then a charge is sent down so that only the capacitor at the specified pixel receives a charge. The capacitor holds the charge till the next refresh cycle. By carefully controlling the amount of voltage supplied, one can allow only some light to pass through. By doing this in very exact and very small increments, LCDs can create a gray scale.

To show colour, an LCD must have

three subpixels with red, green and blue filters to create each colour pixel. By careful control and variation of the voltage applied, the intensity of each subpixel can be varied over 256 shades. Combining all the subpixels gives a possible palette of 16.8 million colours (256 shades of red x 256 shades of green x 256 shades of blue). A typical 12-inch LCD would support resolutions up to 1024x768, which translates into 2,359,296 transistors (1,024 columns x 768 rows x 3 subpixels). Now try doing the same math for a 24-inch TFT.

an image on a 15-inch TFT looks larger than the same image on a 15-inch CRT. This is because a 15-inch TFT directly translates into 15-inch viewable area unlike CRTs where the viewable sizes are lower (for a 17-inch CRT, the viewable size may vary from 15.5 inch to 16 inch).

Hence, the launch of Samsung's SyncMaster 240T in the Indian market is something of a first. With a 24-inch screen size, it is the largest available TFT in the market. Insanely expensive too! It costs over Rs 5,00,000. It is primarily targeted at video editors and CAD/CAM professionals who need larger working display areas.

